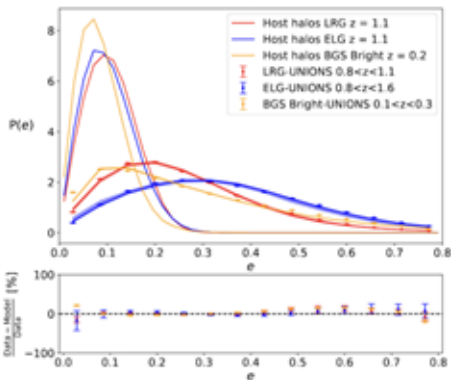


Inferring the 3D shapes of galaxies inside halos with UNIONS

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- 1 **Context:** galaxy-halo connection in 3D for intrinsic alignments
- 2 **Methodology:** simulation-based inference using a forward model of 3D shapes of galaxies in halos
- 3 **Results:** constraints on 3D shapes for 3 galaxy samples from the distributions of ellipticities of galaxies and halos

$C_{A_2} = 0.48 \pm 0.27$
 $C_{A_1} = 0.36 \pm 0.15$
 $C_{A_3} = 0.50 \pm 0.16$

LRG-UNIONS $0.8 < z < 1.1$
ELG-UNIONS $0.8 < z < 1.6$
BGS Bright-UNIONS $0.1 < z < 0.3$



$B_{A_2} = 0.82 \pm 0.11$
 $B_{A_1} = 0.48 \pm 0.11$
 $B_{A_3} = 0.59 \pm 0.19$



Takeaway: we can constrain the distribution of 3D shapes of galaxies from observations, using simulations.